

MESA: Requirements-Oriented Review of AI Benchmarks as Measurement Instruments

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Abstract—Requirements Engineering (RE) for AI increasingly relies on benchmark scores to justify model selection, safety claims, and procurement decisions. Yet AI benchmarks are rarely reviewed as measurement instruments: their intended constructs, scoring assumptions, evidence limits, and validity arguments are often left implicit. This paper introduces the Meta-Evaluation Standard for AI Benchmarks (MESA), a requirements-oriented framework for auditing AI benchmarks before their scores are used to support capability claims. MESA adapts the European Federation of Psychologists’ Associations (EFPA) test review model to the field of AI benchmarks, and is divided into two parts. Part A is descriptive: it records the benchmark’s provenance, task design, target systems, scoring procedure, reporting outputs, and lifecycle materials. Part B is evaluative: it assesses whether the available evidence supports the intended interpretation of benchmark scores, including reliability, validity, contamination risk, comparability, maintenance, and claim proportionality. We demonstrate MESA on three contrasting benchmarks: Humanity’s Last Exam (HLE), ARC-AGI-2, and DesignQA. The reviews show that HLE is best interpreted as a difficult closed-ended academic question-answering benchmark, not as a standalone basis for high-stakes ranking or broad intelligence claims. ARC-AGI-2 is best interpreted as an exact-grid abstraction benchmark under specified evaluation conditions, not as a standalone test of general intelligence. DesignQA is best interpreted as a document-grounded engineering-requirements question-answering benchmark, with limited claims but important gaps in uncertainty reporting, licensing, human baselines, and test-set governance. MESA provides an open-source and replicable quality standard for specifying, auditing, and validating benchmark-based claims in AI RE.

Index Terms—Requirements Engineering for AI, AI benchmarks, benchmark auditing, score interpretation, measurement instruments, construct validity, meta-evaluation, EFPA Test Review Model

I. INTRODUCTION

Benchmarks are one of the main mechanisms by which AI systems are compared, ranked, marketed, and interpreted. In practice, benchmark scores are often treated as direct indicators of capability and reused as evidence for model selection, procurement, safety claims, and public narratives about what AI systems can do. However, a benchmark score is not a capability itself. It is an inference drawn from a designed measurement instrument that operates under assumptions about content, administration, scoring, comparison, and interpretation. Benchmark quality is therefore central to whether AI model capability claims should be trusted.

Recent research has identified multiple weaknesses in AI benchmarking practice, including benchmark contamination (i.e., test items leaking into model training data), prompt sensitivity (where scores vary significantly when prompt wording changes), saturation (i.e., top models scoring near the ceiling, leaving insufficient room to discriminate between them), over-interpretation of leaderboard rankings beyond what the underlying numbers support, unclear definitions of what the benchmark is supposed to measure, weak documentation, and poor reproducibility [1]. These problems threaten construct validity, where a model may score well because the benchmark is narrow, fragile, leaky, or poorly governed rather than because the model is genuinely capable.

This paper argues that AI benchmarks should be reviewed with the same rigor as measurement instruments used to evaluate human cognitive ability. In human assessment, mature disciplines do not only score people on tests; they also review the tests themselves. The EFPA *Test Review Model*, informed by the *Standards for Educational and Psychological Testing*, offers a structured template for examining whether a test’s content, scoring, evidence base, and interpretive claims are justified [2], [3]. AI benchmarking lacks a similarly established review practice, despite the fact that benchmark results increasingly shape research agendas, procurement decisions, public claims about AI capabilities, and policy debates.

The core operational requirement for benchmark use is *traceability*, which reflects that a score interpretation should be traceable from claimed capability to task design, metric, score, comparison basis, and intended use. A score, however, is only as trustworthy as the measurement system that produced it, and that system’s tasks, prompts, access conditions, scoring rules, baselines, and reporting surfaces may be poorly specified, unstable, or misaligned with the capability claim it is being used to support [4], [5], silently propagating undetected measurement weaknesses into safety, fairness, and reliability decisions. This is already observable in software and requirements-engineering practice, where benchmarks that increasingly define agentic-system requirements — scoring whether an agent can resolve repository-level issues, operate a terminal, or solve competitive programming tasks — become the verification-and-validation evidence cited to accept or reject a requirement, yet rarely undergo the same scrutiny themselves.

To address this gap, this paper proposes **MESA** (*Meta-Evaluation Standard for AI Benchmarks*), a requirements-oriented framework for auditing AI benchmarks as measurement instruments. MESA’s role is to help ensure that the test suites supplying model capability evidence are themselves robust, reproducible, and free from execution or contamination bias, making benchmark auditing a prerequisite for—rather than a substitute for—requirements verification. It is not another model-scoring benchmark. Rather, it is a framework for reviewing benchmarks themselves before their scores are used as evidence. MESA draws on three sources: it (1) adapts the EFPA Test Review Model’s instrument description and evaluation, (2) incorporates benchmark-quality concerns [6] including documentation, reproducibility, implementation transparency, and lifecycle maintenance, and (3) applies Cattell-Horn-Carroll theory inspired construct-validity and capability-area perspectives to discipline the inferential chain from phenomenon to task, metric, score, and claim [7], [8].

With this, the paper aims to address three research questions:

- 1) **RQ1:** How can the EFPA Test Review Model be adapted into an RE-oriented quality model for benchmark-based AI capability claims?
- 2) **RQ2:** What benchmark materials and construct validity support are needed for trustworthy AI benchmark score interpretation?
- 3) **RQ3:** What interpretation risks and evidence gaps does MESA reveal across three contrasting AI benchmarks: Humanity’s Last Exam, ARC-AGI-2, and DesignQA?

MESA helps requirements engineers, benchmark users, benchmark maintainers, and AI assurance stakeholders judge which benchmark claims are supported, which are evidence gaps, and how benchmark scores should or should not be used.

II. RELATED WORK

A. AI Benchmarking Practices and Challenges

Concerns across the AI benchmarking literature cluster around two questions: what a benchmark actually measures and how its scores are used, such that a recurring concern is the validity and scope of evaluations [9]. Critiques find that popular benchmarks are often narrow, fragile, or unclear about their target construct, and that headline numbers can reflect saturation or shortcut exploitation rather than the capability they are taken to indicate [1], [5]. Regarding how scores are used, work on leaderboard practices documents systematic over-interpretation and distortion — selective comparison, unreported evaluation conditions, and ranking instability — that make leaderboard positions weaker evidence than their public presentation suggests [4]. Contamination and selective reporting worsen the problem: as benchmarks age and leak into training corpora, memorization inflates scores while competitive pressure incentivizes gaming and cherry-picking [10]. A related failure is benchmark drift, where an instrument is repurposed beyond its original design — as when the Penn

Treebank [11], built for syntactic parsing, was later reused as a perplexity benchmark in ways that detached the metric from its intended meaning.

In response, the community has emphasized documentation and standardization. Datasheets for datasets establish structured reporting of provenance, intended use, and limitations [?], while others score benchmarks against best-practice criteria spanning design, implementation, documentation, and maintenance [6]. These efforts are valuable but largely *developer-facing* and artifact-level: they help creators build and document better benchmarks rather than help a downstream reviewer decide what an existing benchmark’s scores are permitted to mean. MESA builds on and expands this literature by unifying these threads into a single, *reviewer-facing* instrument-review template that separates factual description from evaluative judgment and disciplines the inferential chain from phenomenon to task to metric to claim. The gap MESA addresses is that AI benchmarks are widely used as measurement instruments, yet are rarely reviewed with a unified structure that ties every judgment to evidence, missing evidence, and a bounded statement of supported and non-supported use.

B. Benchmarking standards across the fields

Beyond the AI benchmarking literature, MESA is informed by review and measurement practices from adjacent fields — requirements engineering, psychometric test review, construct-validity theory, and cognitive-ability research — and we identify common themes regarding what makes a measurement instrument trustworthy. Where applicable, we adapt these practices into the MESA Test Review Model.

Requirements engineering for AI systems. RE research emphasizes that AI development changes the subjects and practices of requirements work. Habiba et al. [12] provide a systematic mapping of 126 primary studies, showing growth around elicitation and analysis but persistent challenges in specification, validation, and quality requirements. This motivates MESA’s framing of benchmark claims, tasks, scoring rules, and reproducibility materials as requirements-relevant review objects rather than neutral background infrastructure.

Psychometric instrument review. The EFPA Test Review Model [3] evaluates psychological and educational instruments in two parts: a descriptive part (identity, users, administration, scoring, reports) and an evaluative part (rationale, materials, norms, reliability, validity, fair use, and final interpretation), and cautions against mechanically averaging ratings where expert judgment is required. MESA adapts this structure because a benchmark score, like a test score, is not self-justifying: it requires evidence that the instrument supports the meaning users attach to it.

Construct validity and the phenomenon–task–metric–claim chain. Bean et al. [7] supply a psychometric validity lens, framing benchmark validity as a chain from the real-world phenomenon, through the chosen task and scoring metric, to the claim made about the score. This chain maps naturally onto requirements traceability, and MESA uses it to

test whether a score is valid evidence for the requirement-like claim attached to it.

Broad-intelligence and AGI claims. Hendrycks et al. [8] confront the strongest benchmark claims — those equating performance with general intelligence — grounding AGI in Cattell–Horn–Carroll (CHC) theory as a breadth-weighted profile across cognitive domains, and reporting a ‘jagged’ profile in current models that overstates general competence. MESA draws the lesson of claim proportionality: strong performance on one task family is narrow task-level evidence unless a validity argument spans the relevant abilities. MESA does not require every benchmark to be a full cognitive battery and acknowledges that CHC interpretation remains debated [13].

III. METHODOLOGY

This study focuses on framework construction, and demonstrates MESA’s utility through targeted case studies. The objective is to establish a rigorous and replicable review model for auditing benchmark design, materials, score interpretations, and evidence gaps. The case studies are designed to verify that MESA is feasible to apply, comprehensive in coverage, and useful in practice, not to draw generalized conclusions about the quality of AI benchmarking as a field.

Figure 1 summarizes the construction logic, in which the input sources shown in the caption feed the MESA Test Review Model. Table I lists the benchmarks and primary evaluation focus used as the source for the template requirements.

A. Framework Construction

MESA was derived to make the benchmark itself a review object. As illustrated in Figure 1, 15 benchmarks acted as input sources to help form a structural model adapted from the EFPA Test Review Model. The adaptation was functional rather than textual, following four rules: *preserve* fields whose review function transfers directly to AI benchmarking; *rename* fields when the function transfers but where the previous human-centric terminology is misleading; *replace* fields when the original assumes human-assessment conditions incompatible with AI benchmarks; and *add* AI-specific fields where the EFPA model would otherwise overlook a material benchmark dependency. For example, the EFPA test taker maps to the evaluated AI system and the target population to the intended AI systems or model class; psychometric norms expand into technical reference points (baselines, task comparators, chance floors, human or expert references, model cohorts, and criterion thresholds); and digitally generated reports translate into benchmark papers, result files, dashboards, scorecards, and leaderboards. The added fields incorporate modern engineering parameters such as prompt templates, hidden data splits, scorer code execution, model-as-judge behaviors, API runtime conditions, contamination controls, software versioning, and leaderboard governance.

B. Adaptation and Evidence Integration

The structural adaptation followed four foundational rules. First, preserve fields whose underlying review function transfers directly to AI benchmarking. Second, rename fields when

the core function transfers but the human-centric terminology is misleading in the context of AI. Third, replace fields when the original framework assumes human-assessment conditions that are incompatible with AI benchmarks. Fourth, add AI-specific fields where the EFPA model would otherwise overlook a material benchmark dependency. These new additions incorporate modern engineering parameters, including prompt templates, hidden data splits, scorer code execution, model-as-judge behaviors, API runtime conditions, contamination controls, software versioning, leaderboard governance, and more.

AI benchmark-quality literature was then integrated into the EFPA-derived structure. Reuel et al. [6] informed lifecycle, reproducibility, implementation, documentation, maintenance, and contamination dimensions. Bean et al. [7] informed the phenomenon-task-metric-claim chain. AGI and CHC literature informed the review of broad cognitive claims [8].

C. Intended Empirical Validation Strategy

The present study establishes feasibility and coverage through the structured, LLM-assisted case studies described in Section V; it does not yet validate MESA as a measurement instrument. The intended validation strategy centers on human-in-the-loop consensus panels: independent requirements engineers, benchmark developers, model evaluators, and domain experts would apply MESA to a shared benchmark cohort, allowing us to estimate inter-rater agreement on Part B ratings, quantify the effect of reviewer training, and compare reconciled human panels against the multi-LLM pipeline. This planned study is detailed as the primary thread of future work (Section IX).

IV. THE MESA TEST REVIEW MODEL

The MESA Test Review Model treats an AI benchmark as both a requirements object and a source of score-based claims. The review model has two main parts: Part A describes the benchmark and its materials, while Part B evaluates whether the benchmark supports the interpretations users are expected to draw from its scores. Table II summarizes the high-level structure of the full review model.

a) Part A, Description: Part A describes the benchmark before evaluating it. It records the benchmark’s factual identity, classification, scoring procedure, outputs, and access or lifecycle materials, corresponding to Sections 1–5 of Table II. Part A does not assume that the benchmark is fully documented. If a benchmark provides only a paper, leaderboard, dataset, or partial repository, MESA records unavailable descriptive information as an explicit evidence gap. This prevents missing documentation from being silently filled in, and it gives Part B a clear factual basis for distinguishing documented weaknesses from unknowns.

b) Part B, Evaluation: Part B evaluates whether the benchmark supports its intended score interpretation. It preserves EFPA’s evaluative structure, represented in Sections 6–12 of Table II, but adapts those categories to AI benchmark concerns. In particular, MESA foregrounds reliability across

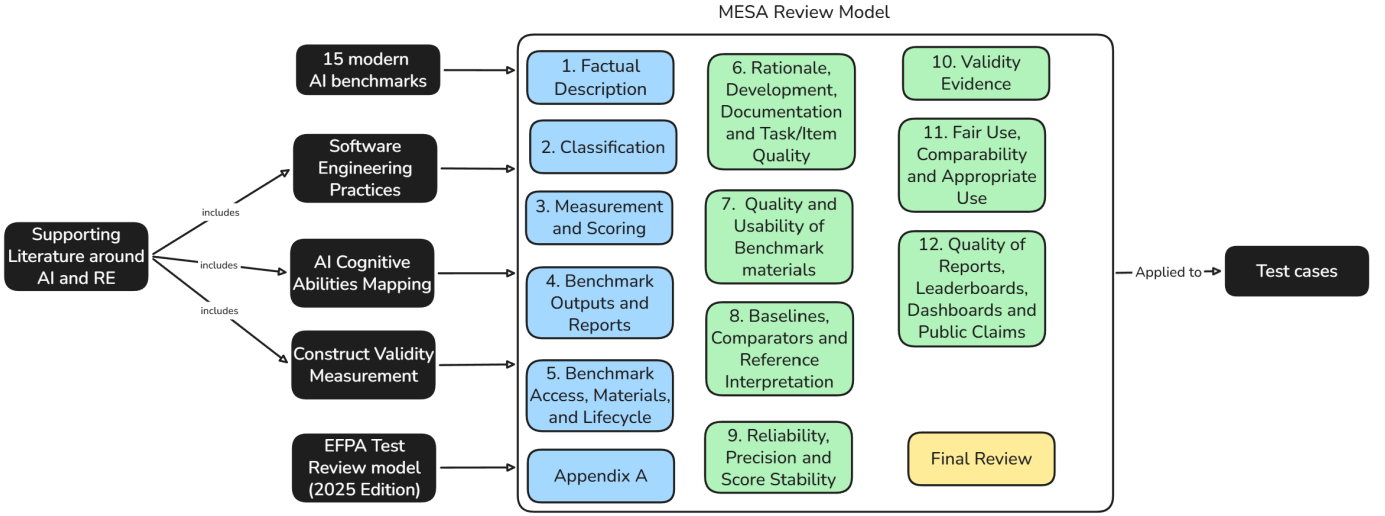


Figure 1. MESA methodology pipeline. Supporting AI and RE literature, a 15-benchmark source inventory, software-engineering practices, AI cognitive-abilities mapping, construct-validity measurement literature, and the EFPA Test Review Model feed the MESA Test Review Model, whose descriptive and evaluative sections are then applied to case studies.

Table I
MODERN AI BENCHMARKS USED AS SOURCE FOR TEMPLATE REQUIREMENTS

#	Benchmark	Primary evaluation focus	#	Benchmark	Primary evaluation focus
1	Humanity’s Last Exam	Academic reasoning	9	Video-MMMU	Knowledge acquisition from videos
2	ARC-AGI-2	Visual reasoning puzzles	10	LiveCodeBench Pro	Competitive coding problems
3	DesignQA	Engineering requirements comprehension	11	SWE-Bench Verified	Agentic coding
4	GPQA Diamond	Scientific knowledge	12	tau2-bench	Agentic tool use
5	AIME 2025	Mathematics	13	Vending-Bench 2	Long-horizon agentic tasks
6	MathArena Apex	Challenging math contest problems	14	MMMLU	Multilingual multiple-choice question answering
7	MMMU-Pro	Multimodal understanding and reasoning	15	Global PIQA	Commonsense reasoning across cultures
8	OmniDocBench 1.5	OCR document analyzing			

runs, prompts, scorers, judges, model versions, and evaluation environments (Section 9). It also expands validity review to include contamination, gameability, and claim proportionality: whether a system can obtain a high score through shortcuts, exposure, formatting, tuning, or benchmark-specific scaffolding rather than the target capability itself (Section 10).

MESA uses EFPA-style ratings of *n/a* and 0–4. The rating *n/a* means that an attribute is not applicable to the reviewed benchmark. A 0 means that the attribute is applicable, but the available information is absent or insufficient for evaluation. A 1 means that evidence exists, but is inadequate for the benchmark’s stated purpose. This distinction is important: missing documentation is an evidence gap, not automatically proof of poor benchmark quality. The final evaluation therefore does more than report ratings. It states what the score can support, what it cannot support, and what conditions must be held fixed for comparisons to be meaningful, such as benchmark version, split, prompt, harness, scorer, comparator cohort, uncertainty, and run conditions. MESA thus replaces mechanical averaging with a traceable account of appropriate score use.

c) Template coherence: The MESA template contains deliberate overlap because the same benchmark artifact can matter for different review functions. For example, a prompt template is first recorded in Part A as benchmark material. It may then be evaluated in Section 7 for usability, Section 9 as a possible source of score instability, Section 10 as a validity threat, and Section 12 as information needed to interpret leaderboard results. This is not redundant as long as each section preserves its role: description, material usability, reliability, validity, fairness, or communication. The same logic applies to other cross-cutting artifacts, including hidden test sets, scorer code, benchmark versions, and contamination controls.

d) MESA as its own measurement object: MESA can also be described using the phenomenon–task–metric–claim chain that it applies to benchmarks. In this meta-evaluation setting, the phenomenon of interest is whether benchmark score interpretations are defensible for Requirements Engineering (RE) use. The task is the structured review process itself: compiling the source inventory, completing Part A descriptions, grounding Part B ratings in evidence, and producing

Table II
HIGH-LEVEL STRUCTURE OF THE MESA TEST REVIEW MODEL

Component	High-level purpose
Part A Section 1. Factual Description	Identifies the review event, benchmark entity, provenance, inspected sources, and access conditions.
Part A Section 2. Classification	Records claimed capability domains, intended systems and users, task families, response modes, conditions, versions, and broad-claim flags.
Part A Section 3. Measurement and Scoring	Describes how model behavior becomes scores, including scoring procedures, metrics, transformations, baselines, uncertainty, and stated score interpretation.
Part A Section 4. Benchmark Outputs and Reports	Records papers, reports, leaderboards, dashboards, result files, and how scores are presented.
Part A Section 5. Benchmark Access, Materials, and Lifecycle	Records repositories, datasets, hidden splits, contamination documentation, reproducibility materials, licenses, maintenance, and versioning.
Appendix A. General Description	Provides a concise non-evaluative summary of the benchmark for readers.
Part B Section 6. Rationale, Development, Documentation, and Task/Item Quality	Evaluates construct definition, development process, documentation, and task quality.
Part B Section 7. Quality and Usability of Benchmark Materials	Evaluates datasets, prompts, rubrics, harnesses, scorers, interfaces, licenses, and usability.
Part B Section 8. Baselines, Comparators, and Reference Interpretation	Evaluates chance, human, expert, model, historical, and criterion reference points.
Part B Section 9. Reliability, Precision, and Score Stability	Evaluates score stability across runs, prompts, scorers, forms, versions, and environments.
Part B Section 10. Validity Evidence	Evaluates whether task design, scoring, contamination controls, and claim proportionality support intended score interpretations.
Part B Section 11. Fair Use, Comparability, and Appropriate Use	Evaluates responsible comparison across systems, access modes, compute budgets, languages, modalities, and intended uses.
Part B Section 12. Quality of Reports, Leaderboards, Dashboards, and Public Claims	Evaluates whether reporting surfaces communicate scores, conditions, uncertainty, caveats, and claims responsibly.
Final Evaluation	Integrates supported interpretations, ratings, evidence gaps, cautions, recommendations, and non-use cases.

a final judgment. The metric consists of EFPA-style ratings, *n/a* and 0–4, together with qualitative evidence, missing-evidence notes, and reviewer cautions. MESA intentionally avoids reducing these judgments to a single aggregate quality score. Its claim is narrower: it supports source-grounded statements about whether a benchmark score can carry a stated interpretation. It does not produce a generic verdict on the benchmark as a whole.

e) Scope of MESA judgments: Two boundary conditions follow from this framing. First, MESA reviews the benchmark as a measurement instrument, not the model being evaluated. The meaning of a benchmark score still depends on model-side evidence. For example, the same high score has different implications if the model may have seen the test set during training, used benchmark-specific scaffolding, or was evaluated under a different prompt or retrieval setup. A complete capability argument must therefore combine a MESA review of the benchmark with model-side evidence, such as training data, prior capabilities, runtime scaffolding, and documentation artifacts, as well as the intended use of the score. MESA supplies a necessary part of that argument, not the whole argument.

Second, the multi-LLM pipeline used in the case studies (Section V) is an assistive drafting and schema-mapping tool, not a replacement for expert judgment. For high-stakes assurance cases, safety certifications, or procurement decisions, MESA reviews should be adjudicated by human expert panels. The framework structures and accelerates the review process; it does not eliminate the need for expert responsibility.

V. CASE STUDY DESIGN

MESA was applied to three benchmarks selected to vary in task format, scoring procedure, and public framing: Humanity’s Last Exam (HLE), ARC-AGI-2, and DesignQA. HLE

is an expert-level academic question-answering benchmark with multiple-choice and short-answer items, including a partially multimodal subset; it uses a model-as-judge procedure to assess answer equivalence [14]. ARC-AGI-2 consists of visual grid-transformation tasks with public tasks and hidden evaluation splits; it is scored by exact cell-level grid matching and is publicly framed around progress toward general intelligence [15]. DesignQA evaluates question answering over an engineering rule document and CAD-derived data; it uses lexical and semantic similarity metrics, including F1, BLEU-2, ROUGE-L, and Sentence-BERT, and is framed more narrowly as an engineering-documentation benchmark rather than an AGI benchmark [16].

To reduce single-reviewer bias, each case followed the multi-LLM pipeline in Figure 2. Two reviewer models (Reviewer 1, ChatGPT 5.5; Reviewer 2, Claude Opus 4.7) ran the same benchmark-reviewer specification on the same per-benchmark inputs and produced MESA drafts without seeing each other’s output. A supervising editor (Gemini 3.1 Pro) then reconciled each discrepancy—in Part A facts, Part B ratings, and Part B rationales—against the sources, and the reconciled review was checked for source grounding, descriptive–evaluative separation, and reviewer cautions before being recorded as the case output. This is a feasibility demonstration of the workflow, not a validation study (Sections VIII and IX).

a) Worked example: For HLE Section 9 (Reliability), Reviewer 1 rated 2 and Reviewer 2 rated 1; the editor set the final rating to 1 using the four-element anchoring model. The **evidence** is a Wald confidence interval in the scoring harness and a 15–18% expert disagreement rate in the paper. The **missing evidence** is run-to-run stability, prompt-sensitivity analysis, and validation of the `o3-mini` judge against human graders, with leaderboard point estimates that omit uncertainty. The

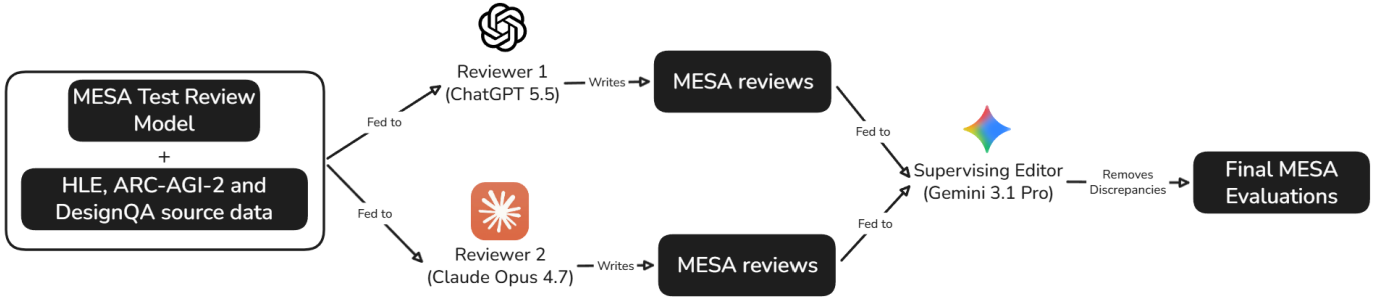


Figure 2. Case study review pipeline. Each benchmark’s information sources (HLE, ARC-AGI-2, or DesignQA) were paired with the MESA Test Review Model and sent to two independent reviewer models. A supervising editor model reconciled the resulting MESA by removing discrepancies to produce final MESA evaluations.

editor rationale is that, for a leaderboard ranking models by thin margins, absent uncertainty and an unvalidated judge are reliability gaps, not missing documentation. The **supported-use and non-use statement** is that HLE results give a coarse performance summary but do not support position-by-position rankings. The anchor is this four-element profile, not the number.

A. Replication Package

The replication package is the open-source MESA repository at <https://github.com/MyLittleFoxxie/MESA-meta-evaluation-standard-for-ai>. It contains the manuscript source and bibliography, the MESA template, the reviews for HLE, ARC-AGI-2, and DesignQA, and utility files.

VI. CASE STUDY RESULTS

A. Humanity’s Last Exam (HLE)

MESA reviewed the static HLE release: 2,500 public questions, a documented private held-out split, the scoring and evaluation code, and the public leaderboards. HLE covers multiple-choice and short-answer formats with partial image-conditioned items, automated grading that uses a separate AI model to decide whether an answer matches the reference (a model-as-judge approach), and metrics including accuracy, category-level performance, and calibration error¹ [14]. Item development was expert-led, the dataset and scoring materials are open, and confidence logging exposes miscalibration. Because the grader is itself an AI model, MESA treats checking that grader against human judgments as core evidence [10].

MESA also finds gaps that bound those scores. The grading model (o3-mini) has not been checked against human judges; the paper reports that experts disagree with each other on 15–18% of items, and an independent chemistry and biology reanalysis estimates errors up to 30% in some text-only subjects [17], which caps how high a score can meaningfully go. The benchmark does not report how much scores change when the same model is run again, and no human baseline was measured under the benchmark’s own

¹Calibration error measures the gap between a model’s stated confidence and its accuracy.

constraints (no internet or external tools), so the ‘expert-level’ framing is unsupported. The public leaderboard² drops the paper’s non-AGI caveats and ranks models by very small score differences, with no confidence interval³ to show whether those differences are real.

MESA therefore supports using HLE to measure performance on closed-ended academic tasks (including some multimodal inputs) under the official prompting setup, and to diagnose overconfidence and miscalibration. It does not support using HLE alone to justify fine-grained leaderboard rankings or ‘expert-equivalent’ claims, nor as evidence of Artificial General Intelligence (AGI), autonomous research, open-ended scientific creativity, general cognitive versatility, or long-horizon planning⁴.

B. ARC-AGI-2

MESA reviewed the ARC-AGI-2 ecosystem: public training and evaluation task files, documented semi-private and private evaluation splits, and public leaderboards. Systems infer a generative rule from a few demonstration input-output pairs and must produce exact output grids for unseen test inputs [15]. The task design uses a fixed format with scoring that counts an answer correct only when every cell of the grid matches, and open public tasks. It includes a human reference point from a 407-participant calibration campaign (515 sessions, 13,405 attempts) requiring at least two independent human solves per task, contamination controls via secure semi-private and private splits backed by an offline Kaggle sandbox, and cost-per-task reporting.

Its weaknesses are statistical and conceptual. The materials frame the benchmark as measuring AGI, psychometric intelligence, and fluid intelligence⁵, yet the task family omits language, world knowledge, memory, planning, social cognition, embodiment, tool use, long-horizon agency, auditory processing, and real-world transfer. With 120 tasks in the

²Public leaderboard available at <https://agi.safe.ai>, accessed 2026-05-18.

³A confidence interval is a range around a reported score that shows how much it could vary by chance; without it, two close scores cannot be reliably told apart.

⁴Long-horizon planning refers to multi-step tasks where an agent plans and executes actions across many consecutive turns.

⁵Fluid intelligence refers to the capacity to solve novel problems without relying on prior knowledge or language skills.

public and semi-private splits, a 1% difference is about one task, yet headline scores carry no confidence interval and no measure of how much scores change when a model is run again, so small ranking differences cannot be separated from chance. The leaderboard also mixes unlabeled runtime environments—the Kaggle offline sandbox (12-hour budget on $4 \times$ NVIDIA L4 GPUs) and hosted APIs with cost-bounded scaffolding⁶—which limits cross-system comparison. A documented 2-versus-3 trial inconsistency with the official pass@2 rule⁷ and absent per-task category labels for the four core challenge types block analysis of whether the sub-categories form a coherent construct.

MESA therefore supports using ARC-AGI-2 to measure exact-grid abstraction and rule-induction under specified split, attempt, verification, and compute-cost conditions, with system-to-system comparison valid only when systems run under identical environment and scaffolding constraints. It does not support standalone claims of AGI preparedness, human-level versatility, or broad psychometric intelligence; it is AGI-relevant only as one narrow component within a larger evaluation portfolio.

C. DesignQA

MESA reviewed the public DesignQA materials [16]: the dataset, the GitHub repository, and a manually updated leaderboard. Multimodal models answer 1,451 question-answer pairs across six subsets (Retrieval, Compilation, Definition, Presence, Dimension, and Functional Performance) drawn from the Formula SAE (FSAE) 2024 rule document and MIT Motorsports CAD data, under a full-context (AllRules) or retrieval-augmented (RAG) condition. It is the most requirements-engineering-focused of the three cases. It uses real source materials, expert authoring with non-author review, an error analysis, and a paper that states its limits and makes no AGI claims.

MESA also finds gaps in reliability, validation, and maintenance. DesignQA reports single numbers with no range showing uncertainty and no repeated runs for the main model comparisons. This matters most for the smaller subsets: Functional Performance contains only 16 items, so one item changes accuracy by 6.25 percentage points. The benchmark is also fully public, and the paper discusses fine-tuning on the Retrieval questions, which would amount to training on released test items. Scoring depends on the exact format and wording of the answer: it requires a strict format, compares against a single reference answer, and uses automatic word-overlap scores (BLEU and ROUGE)⁸, so a correct answer phrased differently can be marked wrong and the reported accuracy can understate true performance. Finally, the inspected materials do not include a human baseline, a license file stating how the

⁶Scaffolding refers to the orchestration code, agentic loops, retrieval systems, and tool-access layers wrapped around the model.

⁷Under pass@2, a task is solved if the model produces the correct grid in at most two attempts.

⁸BLEU and ROUGE are automatic scores that measure how much a model’s answer overlaps, word for word, with a reference answer; they can mark a correct answer wrong when it is worded differently or has a different length.

Table III
CASE STUDY PART B OVERALL RATINGS (0–4)

Part B section	HLE	ARC-AGI-2	DesignQA
6. Rationale, documentation, and task/item quality	2	3	3
7. Benchmark materials and usability	3	3	2
8. Reference points and comparison groups	2	3	2
9. Reliability and precision	1	1	1
10. Validity support	1	2	2
11. Fair use and comparability	2	2	2
12. Reports, leaderboards, and public claims	2	2	2

data may be reused (so reuse rights are unclear), or versioned releases.

MESA therefore supports using DesignQA as a diagnostic benchmark for multimodal models on Formula SAE requirements QA, especially when results are reported by subset and the context condition is held fixed. Small subset differences, roughly below 10–15 percentage points, should be treated cautiously unless uncertainty evidence is added. The review does not support fine-grained model ranking, direct comparison by the simple-average score, procurement use, or broad claims about engineering competence. It also cautions against directly comparing AllRules and RAG results: under AllRules the model is given the full rule document, while under RAG it must retrieve the relevant parts itself, so the two conditions test different things.

D. Overall ratings

Table III reports the reconciled Part B section ratings. These ratings are evidence-grounded reviewer judgments, not arithmetic averages of item scores, benchmark leaderboard positions, or model performance rankings.

Across the three benchmarks, the lowest ratings are in reliability and validity (Sections 9–10). This pattern does not mainly reflect poor benchmark construction. Instead, it reflects missing measurement evidence: limited uncertainty reporting, weak score-stability evidence, and incomplete validation of what score differences should mean.

The reasons differ by benchmark. For HLE and ARC-AGI-2, the main additional concern is claim proportionality: both benchmarks support narrower task-level interpretations than their public framing often suggests. DesignQA is more cautious in its stated claims, but its ratings are limited by a fully public and therefore tunable test set, unclear licensing, and the absence of human or expert baselines.

Overall, MESA distinguishes between supported task-level uses of each benchmark and broader interpretations that would require additional reliability, validity, and governance evidence.

VII. DISCUSSION

MESA reframes benchmark use as a requirements traceability problem. If a project requirement states that an AI system must reason reliably, answer expert-level questions or

act autonomously, a benchmark score should not be treated as direct proof of an AI model’s capability until the benchmark’s target phenomenon, task sample, metric, execution conditions, baselines, uncertainty, and reporting claims have been reviewed. MESA supplies a structure for that review.

For benchmark selection and acceptance criteria, MESA helps requirements engineers compare not only model scores but the materials and validity support behind them: a benchmark with clear materials, uncertainty reporting, scorer validation, contamination controls, and bounded claims may serve assurance better than a more visible one with ambiguous interpretation. It encourages teams to specify the benchmark version, split, prompt or harness, scorer, threshold, comparator, and non-use caveats instead of naming only a leaderboard score.

This matters most for trustworthy AI, where fairness, robustness, reliability, and capability requirements increasingly depend on benchmark scores in both procurement and emerging regulatory frameworks. MESA asks whether those tests are reproducible, comparable, accessible, maintained, and valid for the intended claim. By making the gap between benchmark scope and capability claims explicit and traceable, it functions as a requirements-level quality gate: not a replacement for domain-specific safety analysis, but a structured precondition for treating any benchmark-based result as trustworthy evidence.

Answering RQ2, the three test applications converged on a minimum set of criteria before benchmark scores can be responsibly reused as RE evidence: a precisely defined target phenomenon with explicit non-use caveats; transparent scoring rules backed by empirical model-as-judge validation; chance, human, and expert reference points under identical evaluation conditions; published uncertainty metrics (confidence intervals, run-to-run variance, prompt sensitivity); and contamination controls, version tracking, and bounded reporting surfaces that show core limitations next to headline scores. An instrument lacking these elements can still be useful, but its outputs must be read as task-level.

MESA’s main contribution is the disciplined separation of benchmark description from benchmark evaluation. In ordinary discourse, descriptions such as hard, new, broad, or popular quietly become quality judgments; MESA blocks that shortcut by asking first what the benchmark claims, how it operates, and what materials and score interpretations it offers, and only then whether the evidence is sufficient. This guards against two errors: unsupported criticism, where missing documentation is taken as proof a benchmark is poor (MESA treats it as an evidence gap whose weight depends on intended use), and unsupported endorsement, where difficulty, popularity, or leaderboard visibility is taken as proof of validity (MESA requires a validity argument connecting phenomenon, task, metric, and claim).

Beyond individual audits, MESA contributes a reusable vocabulary for RE practice. The concepts of evidence gaps, claim-proportionality risks, supported-use and non-use statements, and score-condition specifications give requirements

engineers precise language to document what a benchmark can and cannot certify. This vocabulary can be embedded directly into requirements specifications, procurement criteria, system assurance cases, and AI governance documentation — making benchmark-based claims first-class, traceable artifacts in the RE process rather than informal footnotes.

VIII. LIMITATIONS AND THREATS TO VALIDITY

A. Construct Validity

Construct threats concern whether MESA actually measures the quality of an AI benchmark as a measurement instrument. We mitigate this risk by grounding MESA’s structure in the EFPA Test Review Model [3], refined over decades of educational and psychological testing, and by integrating concrete software-engineering criteria from BetterBench [6] and workload-relevance principles from Alonso and Church [9]. MESA nonetheless inherits the contested boundaries of these sources; for example, the mapping from psychometric categories to benchmark review involves judgment that another adaptation might make differently.

B. Internal Validity

Internal threats concern the subjectivity of MESA’s categorical scoring—a reviewer might rate Section 9 (reliability) differently depending on their tolerance for variance. MESA constrains this by requiring every Part B rating to be justified by documented evidence or flagged as an explicit ‘unknown’ gap; ratings remain judgment-based by design, matching EFPA’s logic. A second threat is generation bias from the LLM-drafted pilot reviews, mitigated by strict textual-grounding rules, independent dual drafting, and human-inspectable reconciliation—though the reconciled outputs may still share LLM blind spots, making the pipeline a feasibility demonstration rather than a substitute for human adjudication.

C. External Validity

External threats concern generalization. MESA was applied to three benchmarks chosen for contrast—HLE, ARC-AGI-2, and DesignQA— and based of 15 benchmarks from widespread fields. While this is a very big span of possibilities, the ever-growing field of AI will no doubt lead to benchmarks that might fall out of the MESA’s current scope.

D. Open-Source Artifact

The MESA model is maintained as an open-source Markdown document in the public repository, so this paper captures a snapshot of a framework that will continue to evolve; future applications should cite or pin the repository commit or access date used for a given review.

Despite these threats, MESA offers an immediate, practical contribution: an open-source, replicable artifact that requirements engineers can apply today, letting limitations be addressed incrementally as they surface, without invalidating existing reviews.

IX. FUTURE WORK

Future work will validate MESA as a review instrument through multi-reviewer studies, inter-rater agreement analysis, structured reviewer training, and refinement of ambiguous rating anchors. The review process should also be evaluated with benchmark maintainers, requirements engineers, model evaluators, psychometric experts, and downstream users. A complementary line of work will validate the multi-LLM review pipeline itself: comparing reconciled LLM outputs against expert human reviewer panels, measuring LLM-reviewer agreement, analyzing sensitivity to the choice of reviewer and editor models, and characterizing the stability of MESA judgments under prompt perturbations.

A second direction is tool support. A full MESA review is intentionally detailed, but many RE workflows need a compact scorecard that preserves evidence without hiding caveats. Future tooling could generate a benchmark requirements profile, supported-interpretation statement, evidence-gap list, and non-use statements from the full review.

Finally, MESA should be applied to a larger benchmark sample. The aim should not be to rank benchmarks mechanically, but to identify recurring weaknesses in benchmark documentation, scoring validity, reliability, contamination controls, maintenance, and public communication.

X. CONCLUSION

AI benchmark scores increasingly function as evidence in Requirements Engineering for AI systems, influencing model selection, acceptance criteria, assurance arguments, and public claims. MESA adapts the EFPA Test Review Model into a requirements-oriented review framework that preserves EFPA’s rule of describing the benchmark before judging it, while integrating AI-specific concerns from benchmark-quality, construct-validity, and broad-intelligence literature. By asking reviewers to document benchmark facts before assigning evaluative ratings and to tie every judgment to evidence, missing evidence, and supported-use and non-use statements, MESA makes benchmark-based claims auditable rather than rhetorical. The HLE, ARC-AGI-2, and DesignQA case studies demonstrate that benchmark scores support narrower interpretations than their public framing or default use suggests. Making that distinction explicit, and traceable from phenomenon to task to metric to claim, is what an RE-oriented benchmark review framework should provide.

For the trustworthy AI requirements engineering community specifically, MESA offers something more than a review checklist: it re-frames benchmark use as a requirements traceability problem. A trustworthiness claim backed by a benchmark score is only as strong as the measurement evidence behind that score. By making that evidence (and its gaps) visible, structured, open-source, and replicable, MESA equips requirements engineers, AI assurance practitioners, and policymakers with the tools to demand, and demonstrate, the evidentiary standard that responsible AI deployment requires.

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